Ch 5.1.3-5: More Cross-validation

Lecture 11 - CMSE 381

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Mon, Feb 14, 2022

Announcements

Last time:

- Lab on Classification

Announcements:

- Fifth homework due Friday! Covers:
  - Fri 2/11 Lab
  - Mon 2/14 Lecture
  - Weds 2/16 Lecture
- Office hours - ONLINE

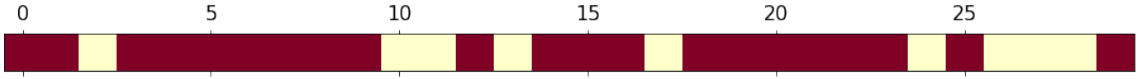
Covered in this lecture

- k-fold CV
- CV for classification

Section 1

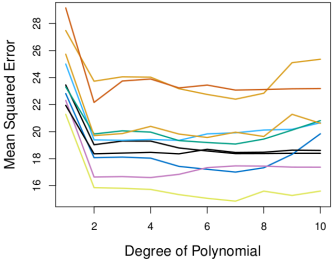
Last time

Validation set approach

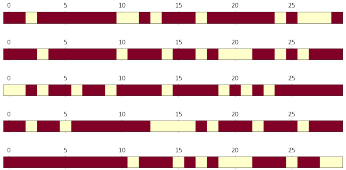


- Divide randomly into two parts:
  - Training set
  - Validation/Hold-out/Testing set
- Fit model on training set
- Use fitted model to predict response for observations in the test set
- Evaluate quality (e.g. MSE)

Problems



Ex. Predict `mpg` using `horsepower`



- Highly variable results, no consensus about the error
- Tends to overestimate test error rate

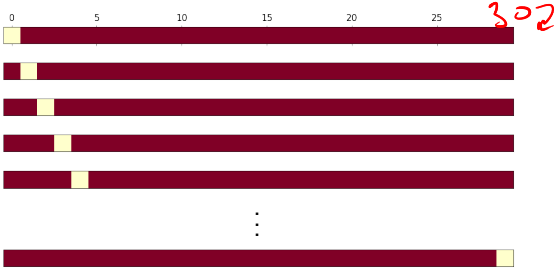
Leave One Out CV (LOOCV)

- Remove (x_1, y_1) for testing.
- Train the model on $n-1$ points: $\{(x_2, y_2), \cdots, (x_n, y_n)\}$
- Calculate $\mathrm{MSE}_1 = (y_1 - \hat{y}_1^{\cancel{1}})^{2}$

30+

- Remove (x_2, y_2) for testing.
- Train the model on $n-1$ points: $\{(x_1, y_1), (x_3, y_3), \cdots, (x_n, y_n)\}$
- Calculate $\mathrm{MSE}_2 = (y_2 - \hat{y}_2^{\cancel{2}})^{2}$

- Rinse and repeat



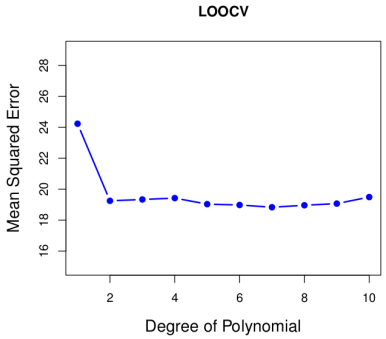
Return the score:

$$CV_{(n)} = \frac{1}{n} \sum_{i=1}^{n} \mathrm{MSE}_i$$

Score each time

times trained

Pros and Cons



No randomness

- No variance
- Higher computation cost

train model n times!

Speeding up LOOCV

Warning: This only works for least squares linear or polynomial regression.

$$h_i = \frac{1}{n} + \frac{(x_i - \overline{x})^2}{\sum_{j=1}^{n}(x_j - \overline{x})^2}$$

leverage

$$\frac{1}{n}\sum_{i=1}^{n} \mathrm{MSE}_i = CV_{(n)} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{y_i - \hat{y}_i}{1 - h_i}\right)^2$$

fit model once on all data

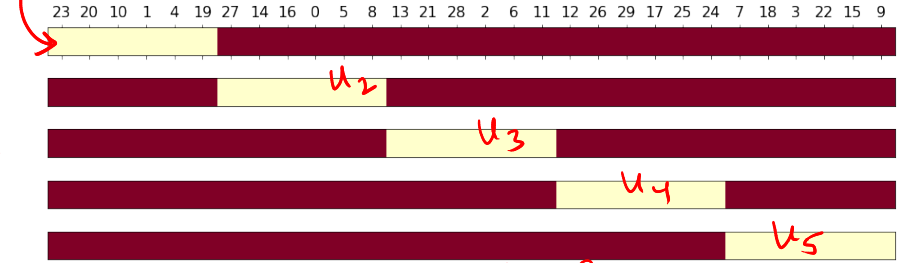
Computational Shortcut

Section 2

k-Fold CV

The idea

randomly permute data



$k=5$

k-folds

Divide data into k piles

Mathy version (Regression)

31 pts
$k = 5$
either 6 or 7 pts / fold

- Randomly split data into k-groups (folds)
- Approximately equal sized. For the sake of notation, say each set has ℓ points

- Remove ith fold U_i and reserve for testing.
- Train the model on remaining points (x_i, y_i)
- Calculate $\mathrm{MSE}_i = \frac{1}{\ell} \sum_{(x_j, y_j) \in U_i} (y_j - \hat{y}_j)^2$ $\hat{y}_i$

# of pts in U_i ↑ data pts in fold

- Rinse and repeat

Return

$$CV_{(k)} = \frac{1}{k} \sum_{i=1}^{k} \mathrm{MSE}_i$$

# of folds

$k = n \Rightarrow$ LOOCV

Pros and Cons

Pros:

Cheaper computation than LOOCV

$k=5$ or $k=10$

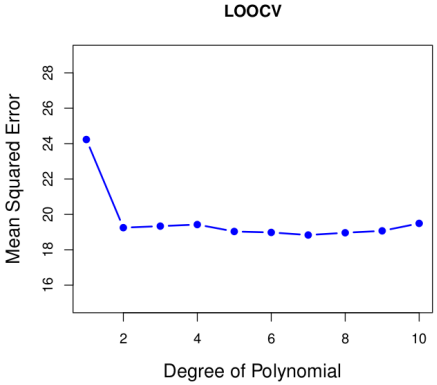
Cons:

There is randomness

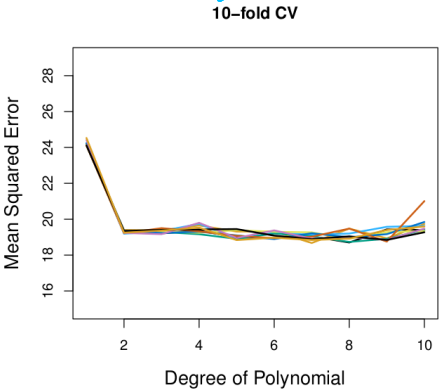
But

Lower variability than Validation

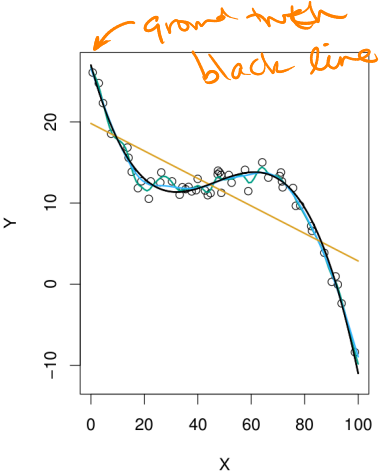
Comparison to LOOCV

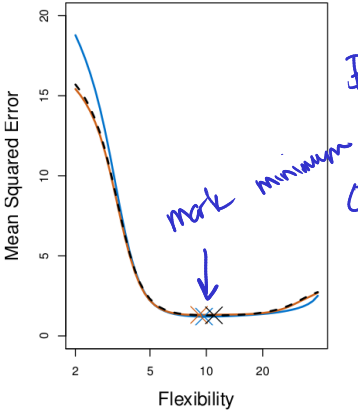


k=10



Comparison with simulated data: Ex 3



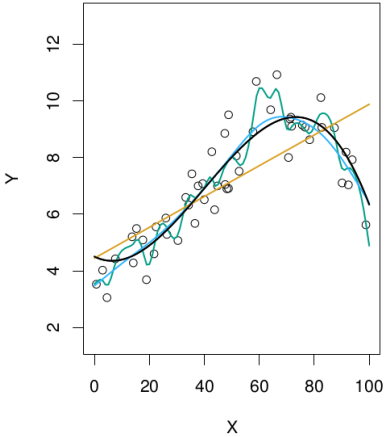


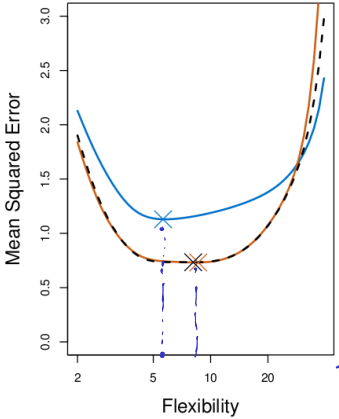
Blue: true test MSE

Black dashed: LOOCV test estimate

Orange line: 10-fold CV test estimate

Comparison with simulated data: Ex 1

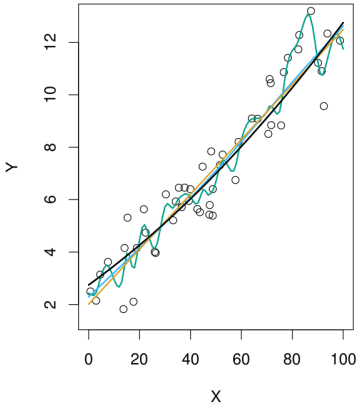


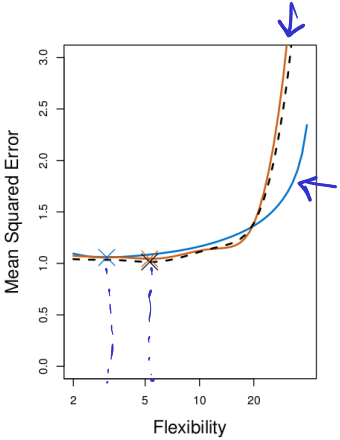


LOOCV, 10-fold both underestimates for true test error

The degrees of freedom is about right

Comparison with simulated data: Ex 2





Close for low flexibility models then over estimates.

Takeaways from the examples

- LOOCV and 10-fold CV similar to each other
- If you only care about min pt in test MSE, good job of approximating even if the MSE is off

Bias-Variance Tradeoff: Bias

$y = f(x) + \varepsilon$

garbage in garbage out

$$E(y_0 - \hat{f}(x_0))^2 = \mathrm{Var}(\hat{f}(x_0)) + \left[\mathrm{Bias}(\hat{f}(x_0))\right]^2 + \mathrm{Var}(\varepsilon)$$

high bias

small

Validation set over estimate test error

↳ used small subset of data to train

k-fold CV medium level of bias

large

LOOCV approximately unbiased estimate

↳ uses most of data to train

low bias

Bias-Variance Tradeoff: Variance

Avg of correlated #'s ⇒ higher variance

Empirically $k=5$, $k=10$

higher variance

$$E(y_0 - \hat{f}(x_0))^2 = \mathrm{Var}(\hat{f}(x_0)) + \left[\mathrm{Bias}(\hat{f}(x_0))\right]^2 + \mathrm{Var}(\varepsilon)$$

LOOCV higher variance

- avg of outputs of n fitted models almost all observations are identical
  → high correlation b/t scores

k-fold CV

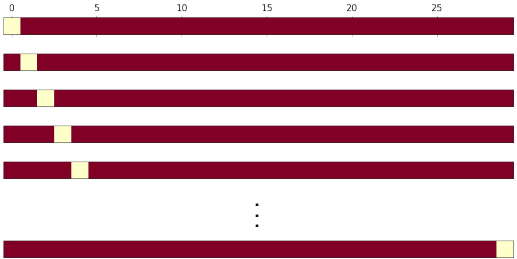
- less correlation b/c less overlap

Section 3

CV for Classification

Setup: LOOCV

- Remove ith point (x_i, y_i) and reserve for testing.
- Train the model on remaining points
- Calculate $\underbrace{\mathrm{Err}_i = \mathrm{I}(y_j \neq \hat{y}_j)}$
- Rinse and repeat



Return

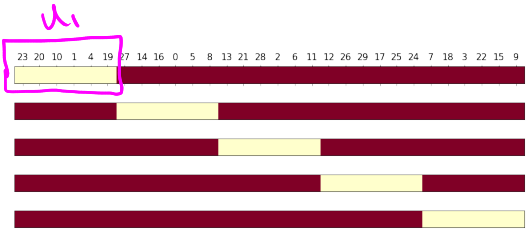
$$CV_{(n)} = \frac{1}{n}\boxed{\sum_{i=1}^{n} \mathrm{Err}_i}$$

count of # times wrong

$$I(y_j \neq \hat{y}_j) = \begin{cases} 1 & y_j \neq \hat{y}_j \\ 0 & y_j = \hat{y}_j \end{cases}$$

Setup: k-fold

- Randomly split data into k-groups (folds)
- Approximately equal sized. For the sake of notation, say each set has ℓ points



- Remove ith fold U_i and reserve for testing.
- Train the model on remaining points
- Calculate $\mathrm{Err}_i = \frac{1}{\ell} \sum_{(x_j, y_j) \in U_i} \mathrm{I}(y_j \neq \hat{y}_j)$

$\ell = \#$ pts in U_i

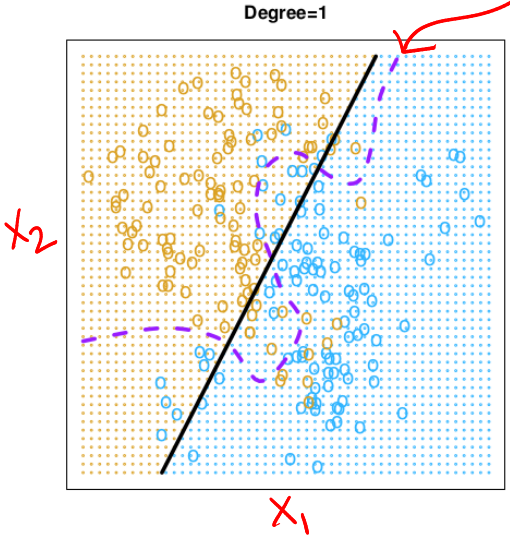
- Rinse and repeat

Return

$$CV_{(k)} = \frac{1}{k} \sum_{i=1}^{k} \mathrm{Err}_i$$

Example on simulated data: Linear

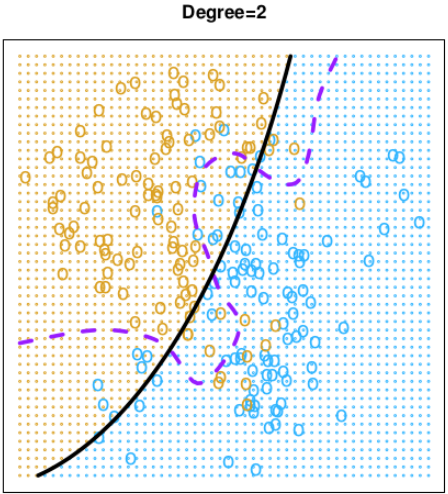
Ground truth



- Purple: Bayes decision boundary.
  - Error rate: 0.133
- Black: Logistic regression
  - $\log(p/(1-p)) = \beta_0 + \beta_1 X_1 + \beta_2 X_2$
  - Error rate: 0.201

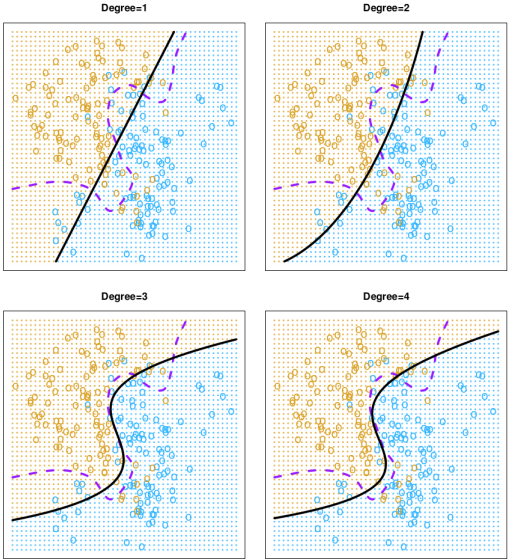
Can we up the degree of the model and do better?

Example on simulated data: Quadratic logistic regression



- Purple: Bayes decision boundary.
  - Error rate: 0.133
- Black: Logistic regression
  - $\log(p/(1-p)) = \beta_0 + \beta_1 X_1 + \beta_2 X_1^2 + \beta_3 X_2 + \beta_4 X_2^2$
  - Error rate: 0.197

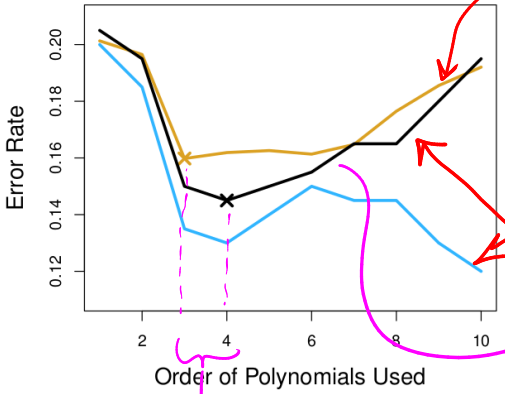
Example on simulated data: all the polynomials!



- Purple: Bayes decision boundary.
  - Error rate: 0.133
- Black: Logistic regression
  - Deg 1 Error rate: 0.201
  - Deg 2 Error rate: 0.197
  - Deg 3 Error rate: 0.160
  - Deg 4 Error rate: 0.162

How to pick b/t models?

Decide degree based on CV



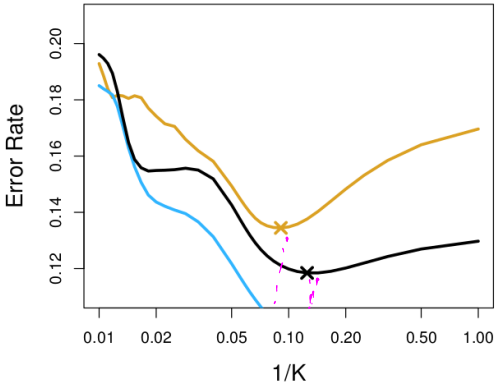
- Test error (brown)
- Training error (blue)
- 10-fold CV error (black)

ground truth

Still have charictanstic U-shape

Good guess for degrees of freedom

Similar game for KNN



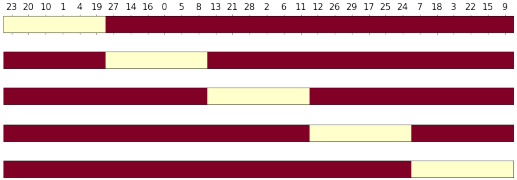
- Test error (brown)
- Training error (blue)
- 10-fold CV error (black)

$\frac{1}{k} = 0.09 = \frac{9}{100}$

$k = \frac{100}{9} \approx 11$ ish

TL;DR

k-fold CV



$$CV_{(k)} = \frac{1}{k}\sum_{i=1}^{k} \mathrm{MSE}_i$$

Use $k = 5$ or 10 usually

Summation

$$\frac{1}{k}\sum_{y_j \in U_i} I(y_j \neq \hat{y}_j)$$

k-fold CV for classification

$$\mathrm{Err}_i = \mathrm{I}(y_j \neq \hat{y}_j)$$

$$CV_{(k)} = \frac{1}{k}\sum_{i=1}^{k} \mathrm{Err}_i$$

Next time

- Weds:
  - 5.2 Bootstrap
- Fri:
  - Lab on Classifiers
  - HW #5 due. Covers:
    - 2/11: Classification Lab
    - 2/14: 5.1.3-5.1.5 k-fold CV; CV for classification
    - 2/16: 5.2 Bootstrap

Next week:

- Mon 2/21: Guest Lecture, Dr. Yuying Xie → In-person
- Weds 2/~~16~~ 23: Online only lecture, 6.1 Subset Selection
- Fri 2/~~18~~ 25: Online only lab, 6.5.1 Subset Selection Methods

Ran very (20 ish minutes)
short. Not honestly
sure what to add. Guying's
Slides have something about
GWAS that maybe could
be used?